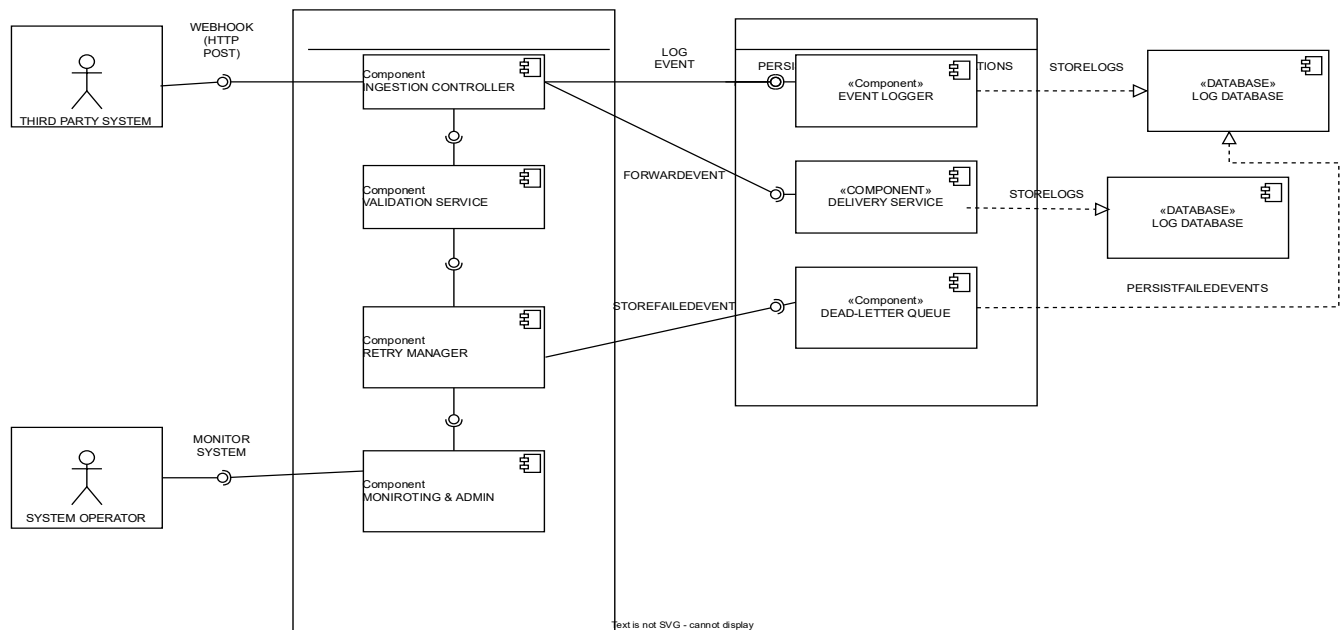


SE LAB 3

NAME: M MAHESH

SRN: PES1UG24AM438

1. DIAGRAM



Architecture Selection – Webhook Ingestion & Retry Mechanism Hub

We chose Microservices Architecture for the Webhook Ingestion & Retry Mechanism Hub.

Microservices architecture is appropriate because the system contains multiple independent responsibilities including webhook ingestion, validation, event logging, delivery, retry handling, and dead-letter persistence. These responsibilities can be implemented as separate components/services with clearly defined interfaces.

Reason 1 – Scalability and Performance:

The system must support throughput of up to **2,500 requests per second** while maintaining high reliability. Microservices allow the Webhook Ingestion Service to be scaled independently when the number of incoming requests increases, without requiring the entire system to be scaled.

Reason 2 – Fault Isolation and Reliability:

Webhook delivery failures should not cause the entire system to fail. The Delivery Service can pass failed events to the Retry Manager, which performs exponential-backoff retries of **1s, 2s, 4s and 8s**. Events that still fail can be persisted in the Dead-Letter Queue instead of being dropped. This improves fault isolation and system resilience.

Security Advantage:

The webhook ingestion and internal processing services can be separated from the external third-party systems through defined interfaces. Validation of incoming webhook data can occur before events are forwarded to internal services, reducing the risk of invalid or malicious payloads reaching internal endpoints.

Performance Benefit:

Independent services allow high-load operations such as webhook ingestion and delivery processing to be scaled separately. This helps the system handle high request volumes efficiently and prevents a heavily loaded component from unnecessarily affecting other components.

Conclusion:

Microservices architecture provides the required scalability, fault isolation, and independent processing capabilities for a resilient webhook gateway while allowing logging, retry, delivery, and dead-letter handling to remain clearly separated.